



Homogenization of Nonlinear Dissipative Wave Equations with Multiscale Temporal Oscillations

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Abstract. We develop a homogenization theory for nonlinear hyperbolic equations with rapid oscillations in space and multiple microscopic time scales. The model features a semilinear wave equation with oscillatory coefficients and separated nonlinearities: an amplitude-dependent potential and a gradient-dependent forcing term. Under appropriate periodicity and structure conditions, we prove convergence of solutions u_ε to an effective macroscopic model as $\varepsilon \rightarrow 0$. The limit problem consists of a homogenized wave operator with a nonlocal-in-time dissipation operator arising from the fast temporal oscillations. Our analysis uses multiscale convergence techniques adapted for wave propagation with multiple temporal scales, extending homogenization theory to a class of nonlinear hyperbolic problems not previously covered.

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1. Introduction

The propagation of nonlinear waves in heterogeneous media is a fundamental problem in applied mathematics, with critical applications spanning geophysics, materials science, and engineering. Prominent examples include seismic wave analysis in viscoelastic geological formations and the ultrasonic characterization of advanced composites. In such systems, the macroscopic wave behavior emerges from the complex interplay of microscopic heterogeneities and multiple nonlinear mechanisms, often operating across widely separated timescales.

Homogenization theory provides a powerful framework for deriving effective macroscopic descriptions of wave motion in composite media. While

the theory is mature for linear elliptic and parabolic equations [3, 4], its extension to *nonlinear, scalar hyperbolic equations* exhibiting rapid temporal oscillations remains an active area of research. Significant progress has been made through techniques such as two-scale convergence [12, 14]. The foundational work [23] established a multiscale homogenization framework for a class of nonlinear hyperbolic equations, while Flodén and Persson [11] developed a comprehensive approach for homogenizing nonlinear dissipative hyperbolic problems using evolution multiscale convergence. These advances, alongside related studies of hyperbolic-parabolic systems [6, 7] and hyperbolic damped stochastic equations [10], have substantially expanded the scope of modern homogenization. More recently, homogenization of PDEs with hysteresis and memory operators has been addressed in [16–19], and stochastic wave equations with damping have been studied in [8, 9].

However, a key limitation persists in these frameworks: the nonlinearity is often encapsulated within a single, monolithic operator that depends on the solution and its gradient. This approach proves insufficient for many physical systems where distinct mechanisms (such as amplitude-dependent potential terms and gradient-dependent forcing contributions) operate concurrently yet independently, each characterized by its own scaling and physical origin. Homogenizing models where these mechanisms are *separated* and exhibit *multi-scale temporal oscillations* remains an open challenge.

In this work, we introduce a novel homogenization framework designed to address this gap. We study the asymptotic behavior of solutions u_ε to a nonlinear hyperbolic equation with multiple, temporally oscillatory nonlinear terms as the heterogeneity scale parameter $\varepsilon \rightarrow 0$. Our main result is the rigorous derivation of an effective homogenized model, which we prove captures the interplay between the different mechanisms at the macroscale. The effective model reveals that the distinct nonlinear effects do not simply superimpose but can interact in a non-trivial way, leading to an emergent effective structure featuring a nonlocal-in-time operator. In particular, the effective equation exhibits a nonlocal-in-time dissipation operator that captures the memory effects induced by the fast temporal oscillations of the coefficients, thereby characterizing the non-trivial coupling between the amplitude-dependent potential and the gradient-dependent forcing at the macroscale.

Specifically, we consider the following initial-boundary value problem:

$$\left\{ \begin{array}{l} \frac{\partial^2 u_\varepsilon}{\partial t^2} - \nabla \cdot \left(A \left(\frac{x}{\varepsilon} \right) \nabla u_\varepsilon \right) + \eta \left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2} \right) \phi(u_\varepsilon) + \psi \left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}, \nabla u_\varepsilon \right) \\ = f \text{ in } Q = \Omega \times (0, T), \\ u_\varepsilon(x, t) = 0 \quad \text{on } \partial\Omega \times (0, T), \\ u_\varepsilon(x, 0) = \frac{\partial u_\varepsilon}{\partial t}(x, 0) = 0 \quad \text{in } \Omega. \end{array} \right. \quad (1.1)$$

Here, $\Omega \subset \mathbb{R}^N$ is a bounded domain with smooth boundary, $T > 0$, and $f \in L^2(Q)$. The functions η and ψ depend on the *fast spatial variable* $y = x/\varepsilon$ and two *fast time variables*, $\tau = t/\varepsilon$ and $\zeta = t/\varepsilon^2$, which are treated as independent variables within the framework of multiscale convergence. The

coefficient matrix $A(y) = (a_{ij}(y))$ and the nonlinear functions

$$\eta : \mathbb{R}^N \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}, \quad \phi : \mathbb{R} \rightarrow \mathbb{R}, \quad \psi : \mathbb{R}^N \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}$$

satisfy the following hypotheses:

- (H1) Periodicity and regularity:** The functions A , η , and ψ are periodic with period 1 in all their microscopic arguments y, τ, ζ . Moreover, $A \in L^\infty(\mathbb{R}^N)^{N \times N}$, and η, ψ are Carathéodory functions, i.e., measurable in (y, τ, ζ) and continuous in their remaining arguments.
- (H2) Uniform ellipticity and boundedness:** The matrix $A(y)$ is symmetric and satisfies, for some $\alpha > 0$,

$$A(y)\xi \cdot \xi \geq \alpha|\xi|^2 \quad \text{for all } \xi \in \mathbb{R}^N, \quad y \in \mathbb{R}^N.$$

- (H3) Nonlinearity assumptions:** There exist constants $\gamma, c_0, \dots, c_8 > 0$, with $0 < \gamma < \infty$ for $N = 1, 2$ and $0 < \gamma \leq 2/(N - 2)$ for $N \geq 3$, such that for all $y, \tau, \zeta, \lambda, \lambda'$, and μ, ν :

- (i) ϕ is non-decreasing: $(\phi(\mu) - \phi(\nu))(\mu - \nu) \geq 0$ for all $\mu, \nu \in \mathbb{R}$.
- (ii) $0 < \eta_0 \leq \eta(y, \tau, \zeta) \leq c_0$, $|\phi(\mu)| \leq c_1(1 + |\mu|^{\gamma+1})$.
- (iii) $\phi(\mu)\mu \geq c_2|\mu|^{\gamma+2} - c_3$.
- (iv) $|\phi'(\mu)| \leq c_4(1 + |\mu|^\gamma)$ (in the sense of distributions).
- (v) $|\psi(y, \tau, \zeta, \lambda)| \leq c_5(1 + |\lambda|)$.
- (vi) $\psi(y, \tau, \zeta, \lambda) \geq c_6 > 0$ (uniform positive lower bound).
- (vii) ψ is globally Lipschitz in λ : for all $\lambda, \lambda' \in \mathbb{R}^N$,

$$|\psi(y, \tau, \zeta, \lambda) - \psi(y, \tau, \zeta, \lambda')| \leq c_7|\lambda - \lambda'|.$$

The monotonicity of ϕ in (i) and its coercivity in (iii) are essential for the compensated compactness argument used to pass to the limit in the potential term. The Lipschitz continuity of ψ in (vii) is essential for the compensated compactness argument used to prove strong multi-scale convergence of the gradients. The strict positivity in (vi) ensures that the gradient-dependent forcing term ψ is uniformly bounded away from zero, providing a strictly positive background contribution regardless of the gradient magnitude. Together, these assumptions guarantee the well-posedness of (1.1) via standard Faedo–Galerkin and compactness arguments (see Sect. 2), and they are sufficient for our convergence analysis.

A prototypical example satisfying **(H1)–(H3)** is given by:

$$\begin{aligned} \phi(\mu) &= |\mu|^\gamma \mu, \\ \eta(y, \tau, \zeta) &= 1 + \alpha_1 \sin(2\pi\tau) + \alpha_2 \cos(2\pi\zeta), \\ \psi(y, \tau, \zeta, \lambda) &= \beta + \kappa(y) \frac{|\lambda|^2}{1 + |\lambda|^2}, \end{aligned}$$

with constants $\beta > 0$, α_1, α_2 chosen such that $\eta \geq \eta_0 > 0$, and a smooth, periodic function κ satisfying $0 \leq \kappa(y) \leq \kappa_0 < \beta$. The function $\phi(\mu) = |\mu|^\gamma \mu$ is monotone, so **(H3)**(i) holds; its growth and coercivity satisfy (ii)–(iv). The scalar function ψ is bounded below by $\beta - \kappa_0 > 0$, so **(H3)**(vi) holds; since the gradient of $|\lambda|^2/(1 + |\lambda|^2)$ is uniformly bounded, ψ is globally Lipschitz in λ , verifying **(H3)**(vii). The boundedness condition **(H3)**(v) is also immediate.

The model (1.1) represents a significant advance in two key aspects. First, the term $\eta(x/\varepsilon, t/\varepsilon, t/\varepsilon^2) \phi(u_\varepsilon)$ captures a nonlinear potential with *dual-scale temporal oscillations*, enabling the modeling of materials where nonlinear effects arise from mechanisms operating at distinct time scales. The temporal variables $\frac{t}{\varepsilon}$ and $\frac{t}{\varepsilon^2}$ represent two distinct microscopic timescales, allowing the model to capture phenomena evolving at different rates. Their precise physical interpretation depends on the application under consideration. For instance, in viscoelastic polymers, the scale t/ε^2 can capture fast, reversible molecular relaxations (e.g., side-chain motions), while t/ε models slower, irreversible processes associated with glassy transitions or segmental rearrangements. Similarly, in wave propagation through poroelastic media, the two scales can represent the distinct characteristic times of the solid matrix response and the fluid diffusion process. Second, the scalar function $\psi(x/\varepsilon, t/\varepsilon, t/\varepsilon^2, \nabla u_\varepsilon)$ independently models a gradient-dependent forcing contribution, relevant to nonlinear reaction mechanisms in heterogeneous media or state-dependent external fields. This explicit separation allows for a more physically realistic representation of coexisting nonlinear effects than previously possible, as it respects the fundamentally different physical origins of these mechanisms.

Our work makes three fundamental contributions to mathematical homogenization:

1. We develop an *extended homogenization theory* that provides the first rigorous framework for nonlinear hyperbolic equations with multiple fast timescales in separated nonlinearities, advancing the mathematical machinery of evolution multiscale convergence to handle this complex interplay.
2. We introduce a *unified modeling approach* capable of analyzing distinct physical mechanisms within a single cohesive model, providing enhanced flexibility for modeling real-world materials with multiple relaxation processes.
3. We establish *foundational convergence results* that rigorously derive the effective macroscopic model from the multiscale system, creating a theoretical bridge between complex microscale physics and practical macroscale applications in geophysics and materials science. In particular, the effective equation features a nonlocal-in-time dissipation operator that captures the memory effects induced by the fast temporal oscillations of the coefficients, thereby characterizing the non-trivial coupling between the distinct mechanisms at the macroscale.

The paper is structured as follows. Section 2 establishes the functional-analytic framework, proving well-posedness and deriving essential a priori estimates. Section 3 develops the core homogenization framework, identifying the form of the homogenized problem. Finally, Sect. 4 presents our main homogenization theorem, rigorously proving convergence to the effective macroscopic equation.

2. Preliminaries

2.1. Well-Posedness and Functional Setting

The mathematical analysis of problem (1.1) requires a functional framework that accommodates both the hyperbolic wave operator and the separated nonlinear structure. Under assumptions **(H1)**–**(H3)**, with ϕ assumed non-decreasing (monotone) as in **(H3)**(i), the well-posedness theory follows from standard monotone operator theory combined with the Lipschitz perturbation ψ . The monotonicity of ϕ is essential for both the existence of solutions and the compensated compactness arguments employed in our homogenization analysis.

Theorem 2.1. *Under assumptions **(H1)**–**(H3)**, for each fixed $\varepsilon > 0$, there exists a unique solution*

$$u_\varepsilon \in C([0, T]; H_0^1(\Omega)) \cap C^1([0, T]; L^2(\Omega))$$

to the semilinear hyperbolic problem (1.1). Moreover, the solution satisfies the uniform energy estimate

$$\|u_\varepsilon\|_{L^\infty(0, T; H_0^1(\Omega))} + \left\| \frac{\partial u_\varepsilon}{\partial t} \right\|_{L^\infty(0, T; L^2(\Omega))} \leq C \|f\|_{L^2(Q)},$$

where $C > 0$ is independent of ε .

Proof. The existence follows from the Faedo–Galerkin method. For each $m \in \mathbb{N}$, we construct finite-dimensional approximations $u_{\varepsilon, m}$ in the space spanned by the eigenfunctions of the Dirichlet Laplacian. The monotonicity of ϕ and the Lipschitz continuity of ψ yield a system of ordinary differential equations that admits a unique local solution. The uniform energy estimate (proven in Proposition 2.1) provides the necessary a priori bounds to extend the solution globally in time and pass to the limit $m \rightarrow \infty$. Uniqueness is a consequence of the Lipschitz property of ψ and the monotonicity of ϕ , which together imply that the difference of two solutions satisfies a linear-type energy estimate that forces it to vanish. Standard compactness arguments complete the proof. For details on the abstract framework, we refer to [15, 18]. $\square$

Remark 2.1. The regularity properties established in Theorem 2.1, combined with the growth conditions in **(H3)**, ensure that the nonlinear terms

$$\eta^\varepsilon(x, t)\phi(u_\varepsilon(x, t)) \quad \text{and} \quad \psi^\varepsilon(x, t, \nabla u_\varepsilon(x, t))$$

belong to $L^2(Q)$, where we employ the notation

$$\eta^\varepsilon(x, t) = \eta\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}\right), \quad \psi^\varepsilon(x, t, \lambda) = \psi\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}, \lambda\right).$$

Consequently, examining the principal differential operator in (1.1) reveals that the second-order time derivative satisfies

$$\frac{\partial^2 u_\varepsilon}{\partial t^2} \in L^2(0, T; H^{-1}(\Omega)).$$

This regularity profile naturally situates the solution within the Banach space

$$\mathcal{W} := \left\{ v = v(x, t) \in L^\infty(0, T; H_0^1(\Omega)) \cap W^{1,\infty}(0, T; L^2(\Omega)) \right. \\ \left. \left| \frac{\partial^2 v}{\partial t^2} \in L^2(0, T; H^{-1}(\Omega)) \right. \right\},$$

equipped with the norm

$$\|v\|_{\mathcal{W}} := \|v\|_{L^\infty(0, T; H_0^1(\Omega))} + \|\partial_t v\|_{L^\infty(0, T; L^2(\Omega))} + \|\partial_t^2 v\|_{L^2(0, T; H^{-1}(\Omega))}.$$

The embedding properties of the space $\mathcal{W}$ play a crucial role in our homogenization analysis. Under the dimensional constraint $1 < p < \frac{2N}{N-2}$ for $N \geq 3$, the classical Rellich–Kondrachov compactness theorem guarantees that the embedding $H_0^1(\Omega) \hookrightarrow L^p(\Omega)$ is compact. This compactness property propagates to the evolutionary setting through the Aubin–Lions–Simon lemma, ensuring that $\mathcal{W}$ is compactly embedded in $L^p(Q)$. This compact embedding is essential for establishing strong convergence results necessary for passing to the limit in the nonlinear terms.

The homogeneous initial conditions $u_\varepsilon(0) = \frac{\partial u_\varepsilon}{\partial t}(0) = 0$ further restrict the solution to the closed subspace

$$\mathcal{W}_0 := \{v \in \mathcal{W} \mid v(0) = \partial_t v(0) = 0\},$$

which inherits the Banach space structure from $\mathcal{W}$ and serves as the natural solution space for both the original and homogenized problems.

2.2. A Priori Estimates

The following uniform estimates form the analytical foundation for our homogenization analysis, providing the essential bounds required for compactness and convergence arguments.

Proposition 2.1. *Under assumptions **(H1)–(H3)**, the solution u_ε of problem (1.1) satisfies the following estimates uniformly in $\varepsilon > 0$:*

$$\|u_\varepsilon\|_{L^\infty(0, T; H_0^1(\Omega))} \leq C \|f\|_{L^2(Q)}, \quad (2.1)$$

$$\left\| \frac{\partial u_\varepsilon}{\partial t} \right\|_{L^\infty(0, T; L^2(\Omega))} \leq C \|f\|_{L^2(Q)}, \quad (2.2)$$

$$\left\| \frac{\partial^2 u_\varepsilon}{\partial t^2} \right\|_{L^2(0, T; H^{-1}(\Omega))} \leq C \|f\|_{L^2(Q)}, \quad (2.3)$$

$$\|\eta^\varepsilon \phi(u_\varepsilon)\|_{L^2(Q)} \leq C \left(1 + \|f\|_{L^2(Q)}^{\gamma+1} \right), \quad (2.4)$$

$$\|\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon)\|_{L^2(Q)} \leq C (1 + \|f\|_{L^2(Q)}), \quad (2.5)$$

where $p = \gamma + 2$ and the constant $C > 0$ is independent of ε .

Proof. We establish the estimates through energy methods and the structural assumptions. Define the energy functional

$$E_\varepsilon(t) := \frac{1}{2} \left\| \frac{\partial u_\varepsilon}{\partial t} \right\|_{L^2(\Omega)}^2 + \frac{1}{2} \int_\Omega A^\varepsilon \nabla u_\varepsilon \cdot \nabla u_\varepsilon \, dx.$$

By the uniform ellipticity **(H2)**, we have $E_\varepsilon(t) \geq c_0(\|\partial_t u_\varepsilon\|_{L^2}^2 + \|\nabla u_\varepsilon\|_{L^2}^2)$.

Testing equation (1.1) with $\partial_t u_\varepsilon$ and integrating over Ω yields the energy identity

$$\frac{d}{dt} E_\varepsilon(t) + \int_\Omega \eta^\varepsilon \phi(u_\varepsilon) \partial_t u_\varepsilon \, dx + \int_\Omega \psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \partial_t u_\varepsilon \, dx = \int_\Omega f \partial_t u_\varepsilon \, dx.$$

For the ϕ -term, using the boundedness of η **(H3)(i)**, the growth of ϕ **(H3)(i)**, and Young's inequality, we have

$$\begin{aligned} \left| \int_\Omega \eta^\varepsilon \phi(u_\varepsilon) \partial_t u_\varepsilon \, dx \right| &\leq C \int_\Omega (1 + |u_\varepsilon|^{\gamma+1}) |\partial_t u_\varepsilon| \, dx \leq C_\delta (1 + \|u_\varepsilon\|_{L^p(\Omega)}^p) \\ &\quad + \delta \|\partial_t u_\varepsilon\|_{L^2(\Omega)}^2, \end{aligned}$$

where $p = \gamma + 2$ and $\delta > 0$ is a small parameter to be chosen below. Note that this estimate uses only the boundedness of η^ε and avoids any problematic $\partial_t \eta^\varepsilon$ terms.

For the ψ -term, using **(H3)(iv)** and Young's inequality, we obtain

$$\begin{aligned} \left| \int_\Omega \psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \partial_t u_\varepsilon \, dx \right| &\leq c_5 \int_\Omega (1 + |\nabla u_\varepsilon|) |\partial_t u_\varepsilon| \, dx \leq C(1 + \|\nabla u_\varepsilon\|_{L^2}^2) \\ &\quad + \frac{1}{2} \|\partial_t u_\varepsilon\|_{L^2}^2. \end{aligned}$$

For the source term,

$$\int_\Omega f \partial_t u_\varepsilon \, dx \leq \frac{1}{2} \|f\|_{L^2(\Omega)}^2 + \frac{1}{2} \|\partial_t u_\varepsilon\|_{L^2}^2.$$

To control the $\|u_\varepsilon\|_{L^p}^p$ term arising from the ϕ -term, we use the coercivity **(H3)(ii)**. Let $\Phi(s) = \int_0^s \phi(r) \, dr$. By **(H3)(ii)**, there exist constants $c > 0$ and $C > 0$ such that $\Phi(s) \geq c|s|^p - C$. Integrating over Ω gives

$$\int_\Omega \Phi(u_\varepsilon) \, dx \geq c \|u_\varepsilon\|_{L^p}^p - C|\Omega|.$$

Since $\int_\Omega \Phi(u_\varepsilon) \, dx$ is controlled by the energy functional with the potential term, but we defined $E_\varepsilon(t)$ without the potential, we need to include it. Define the augmented energy

$$\mathcal{E}_\varepsilon(t) := E_\varepsilon(t) + \int_\Omega \Phi(u_\varepsilon) \, dx.$$

Then by the coercivity of Φ , we have $\mathcal{E}_\varepsilon(t) \geq c_0(\|\partial_t u_\varepsilon\|_{L^2}^2 + \|\nabla u_\varepsilon\|_{L^2}^2 + \|u_\varepsilon\|_{L^p}^p) - C$. Also, since Φ may not be nonnegative, we have the bound $\int_\Omega \Phi(u_\varepsilon) \, dx \leq \mathcal{E}_\varepsilon(t) + C$.

Now, using the estimate for the ϕ -term and choosing δ sufficiently small, we obtain

$$\frac{d}{dt} \mathcal{E}_\varepsilon(t) \leq C \mathcal{E}_\varepsilon(t) + C(1 + \|f\|_{L^2(\Omega)}^2).$$

Since $u_\varepsilon(0) = \partial_t u_\varepsilon(0) = 0$, we have $\mathcal{E}_\varepsilon(0) = 0$. Gronwall's lemma yields $\mathcal{E}_\varepsilon(t) \leq C(1 + \|f\|_{L^2(Q)}^2)$, which proves (2.1) and (2.2). The coercivity of Φ also gives the additional bound

$$\int_Q |u_\varepsilon|^p \, dx dt \leq C.$$

Using the growth condition **(H3)**(i) and the established bound on u_ε :

$$|\eta^\varepsilon \phi(u_\varepsilon)| \leq c_0 c_1 (1 + |u_\varepsilon|^{\gamma+1}) \in L^2(Q),$$

since $u_\varepsilon \in L^p(Q)$ with $p = \gamma + 2$ by the coercivity of Φ and the energy estimate. This proves (2.4).

Similarly, using **(H3)**(iv) and (2.1):

$$|\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon)| \leq c_5 (1 + |\nabla u_\varepsilon|) \in L^2(Q),$$

which establishes (2.5).

From the equation structure:

$$\begin{aligned} \left\| \frac{\partial^2 u_\varepsilon}{\partial t^2} \right\|_{H^{-1}(\Omega)} &\leq \|\nabla \cdot (A^\varepsilon \nabla u_\varepsilon)\|_{H^{-1}(\Omega)} + \|\eta^\varepsilon \phi(u_\varepsilon)\|_{L^2(\Omega)} \\ &\quad + \|\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon)\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)}. \end{aligned}$$

The uniform ellipticity **(H2)** and estimates (2.1), (2.4), (2.5) imply the bound (2.3). $\square$

Corollary 2.1. *The sequence $\{u_\varepsilon\}$ is uniformly bounded in the energy space $\mathcal{W}$:*

$$\|u_\varepsilon\|_{\mathcal{W}} \leq C.$$

Consequently, $\{u_\varepsilon\}$ is relatively compact in $L^p(Q)$ for $1 \leq p < \frac{2N}{N-2}$ when $N \geq 3$, and for all $1 \leq p < \infty$ when $N = 1, 2$.

Proof. The bound follows directly from Proposition 2.1. The compactness in $L^p(Q)$ is a consequence of the Aubin–Lions–Simon lemma applied to the embedding

$$\mathcal{W} \hookrightarrow L^p(0, T; L^p(\Omega)) \quad \text{compactly,}$$

which follows from the compact embedding $H_0^1(\Omega) \hookrightarrow L^p(\Omega)$ and the boundedness of the time derivative in $L^\infty(0, T; L^2(\Omega))$. $\square$

2.3. Microscopic Bilinear Form and Averaged Nonlinearities

In preparation for the homogenization analysis, we introduce the bilinear form that will capture the combined macroscopic-microscopic gradient energy. For $\mathbf{u} = (u_0, u_1)$ and $\mathbf{v} = (v_0, v_1)$ in $\mathcal{W}_0 \times L^2(Q; \mathcal{V})$, where $\mathcal{V}$ is defined in Sect. 3, we define

$$a_Q(\mathbf{u}, \mathbf{v}) = \sum_{i,j=1}^N \iiint_{Q \times Y \times Z^2} a_{ij}(y) \mathbb{D}_j \mathbf{u} \mathbb{D}_i \mathbf{v} \, dx \, dt \, dy \, d\tau \, d\zeta,$$

where

$$\mathbb{D}_j \mathbf{u} = \frac{\partial u_0}{\partial x_j} + \frac{\partial u_1}{\partial y_j}.$$

We also define the averaged nonlinearities

$$\widetilde{\eta\phi}(u_0) = \iiint_{Y \times Z^2} \eta(y, \tau, \zeta) \phi(u_0) \, dy \, d\tau \, d\zeta,$$

and

$$\widetilde{\psi}(\mathbb{D}\mathbf{u}) = \iiint_{Y \times Z^2} \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u}) \, dy \, d\tau \, d\zeta.$$

These quantities will be used in the proof of Proposition 2.4 below and in the formulation of the homogenized variational problem in Sect. 4.

2.4. Notation and Function Spaces

We conclude this subsection by establishing notational conventions that will be employed throughout the remainder of this work. All functions are assumed to be real-valued, and all function spaces are defined over $\mathbb{R}$ unless explicitly stated otherwise. Let $Y = (0, 1)^N$ denote the unit cell of periodicity in spatial variables, and let $Z = (0, 1)$ represent the unit temporal period. We shall use $Z^2 := Z \times Z$ for the two temporal scales. For a given function space $F(\mathbb{R}^N)$, we adopt the following standard notation:

- $F_{\text{per}}(Y)$: the space of Y -periodic functions in $F_{\text{loc}}(\mathbb{R}^N)$,
- $F_{\#}(Y)$: the subspace of $F_{\text{per}}(Y)$ consisting of functions with zero mean value over Y .

Specific instances of these spaces that will be frequently employed include:

- $\mathcal{D}_{\text{per}}(Y) := \mathcal{C}_{\text{per}}^{\infty}(Y)$: smooth periodic test functions,
- $\mathcal{D}_{\#}(Y)$: smooth periodic test functions with vanishing mean,
- $\mathcal{D}'_{\text{per}}(Y)$: the dual of $\mathcal{D}_{\text{per}}(Y)$, identified with periodic distributions in $\mathcal{D}'(\mathbb{R}^N)$.

2.5. Multiscale Convergence

The presence of multiple scales in our problem necessitates a convergence framework capable of simultaneously capturing macroscopic behavior and microscopic oscillations across both spatial and temporal variables. The decomposition of nonlinear terms into separate components requires careful treatment of their distinct scaling behaviors.

We begin with a fundamental concept: a *fundamental sequence* is any sequence $(\varepsilon_n)_{n \in \mathbb{N}} \subset (0, 1]$ such that $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. More generally, let E be any subset of positive real numbers accumulating at zero.

Our analysis employs a sophisticated notion of multiscale convergence that extends the pioneering works [1, 13] to accommodate the three-scale

structure inherent in our model, featuring two distinct microscopic timescales in addition to spatial oscillations.

Definition 2.2. Let $(u_\varepsilon)_{\varepsilon \in E} \subset L^2(Q)$ be a family of functions. We say that u_ε *multiscale converges* to a limit function $u_0 \in L^2(Q \times Y \times Z^2)$, where $Y = (0, 1)^N$ and $Z = (0, 1)$, if for every test function $v \in L^2(Q; \mathcal{C}_{\text{per}}(Y \times Z^2))$, the following limit holds:

$$\lim_{\varepsilon \rightarrow 0} \int_Q u_\varepsilon(x, t) v^\varepsilon(x, t) \, dx \, dt = \int_{Q \times Y \times Z^2} u_0(x, t, y, \tau, \zeta) v(x, t, y, \tau, \zeta) \, dx \, dt \, dy \, d\tau \, d\zeta,$$

where the oscillating test function is defined by

$$v^\varepsilon(x, t) := v\left(x, t, \frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}\right).$$

This convergence is denoted by

$$u_\varepsilon \xrightarrow{\text{m.s.}} u_0 \quad \text{in } L^2(Q)\text{-weak.}$$

The utility of this convergence concept is underpinned by two fundamental results that form the bedrock of our homogenization methodology.

Proposition 2.3. [2] *Any bounded sequence in $L^2(Q)$ admits a subsequence that multiscale converges to some limit in $L^2(Q \times Y \times Z^2)$.*

Proposition 2.4. *Let $(u_\varepsilon)_{\varepsilon \in E} \subset \mathcal{W}$ be a bounded sequence in the energy space $\mathcal{W}$, where E is a fundamental sequence. Then there exists a subsequence $E' \subset E$ and a pair $\mathbf{u} = (u_0, u_1) \in \mathcal{W} \times L^2(Q \times Z^2; W_{\#}^{1,2}(Y))$ such that, as $\varepsilon \rightarrow 0$ along E' ,*

$$u_\varepsilon \rightharpoonup u_0 \quad \text{in } \mathcal{W}\text{-weak}, \quad (2.6)$$

and for each spatial component $j = 1, \dots, N$,

$$\frac{\partial u_\varepsilon}{\partial x_j} \xrightarrow{\text{m.s.}} \frac{\partial u_0}{\partial x_j} + \frac{\partial u_1}{\partial y_j} \quad \text{in } L^2(Q)\text{-weak}. \quad (2.7)$$

Moreover, the compact embedding of $\mathcal{W}$ into $L^p(Q)$ ensures the strong convergence

$$u_\varepsilon \rightarrow u_0 \quad \text{strongly in } L^p(Q), \quad (2.8)$$

for $1 \leq p < \frac{2N}{N-2}$ (or $1 \leq p < \infty$ for $N = 1, 2$).

Proof. This is a standard result in multiscale convergence theory; see, for example, [1, 2]. The compactness of the sequence in $\mathcal{W}$ follows from Corollary 2.1, and the two-scale decomposition of the gradients follows from the classical two-scale compactness theorem. $\square$

Proposition 2.5. *Under the assumptions (H1)–(H3), let $\{u_\varepsilon\}$ be a sequence satisfying the conclusions of Proposition 2.4. Then, along a subsequence,*

$$\eta^\varepsilon \phi(u_\varepsilon) \xrightarrow{m.s.} \eta(y, \tau, \zeta) \phi(u_0) \quad \text{in } L^2(Q)\text{-weak.} \quad (2.9)$$

Proof. The convergence follows from the strong convergence of u_ε in $L^p(Q)$ (from Corollary 2.1) and the continuity of ϕ with the growth condition **(H3)**(i). Specifically, since $u_\varepsilon \rightarrow u_0$ strongly in $L^p(Q)$ with $p = \gamma + 2$, we have

$$\phi(u_\varepsilon) \rightarrow \phi(u_0) \quad \text{strongly in } L^2(Q)$$

by the dominated convergence theorem (using the growth bound $|\phi(u_\varepsilon)| \leq C(1 + |u_\varepsilon|^{\gamma+1})$ and the uniform bound on u_ε in $L^p(Q)$). This implies the multiscale weak convergence. $\square$

Remark 2.2. The convergence of the gradient-dependent nonlinearity

$$\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \xrightarrow{m.s.} \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u})$$

will be established in Sect. 4 as part of the main homogenization theorem. This requires proving strong multiscale convergence of the gradients, which is a consequence of the compensated compactness argument adapted to our setting with monotone nonlinearities. We defer this proof to Sect. 4, where the full homogenization result is presented.

Remark 2.3. The periodic unfolding technique provides an alternative framework for establishing the multiscale convergence results in Propositions 2.4 and 2.5. This approach offers technical advantages in handling the separated nonlinearities and will be employed in the detailed proofs of our main convergence results in Sect. 4. For a comprehensive treatment of the unfolding method, we refer to [5].

3. Homogenization Framework and Preliminary Results

This section establishes the fundamental mathematical framework and preliminary results that underpin our homogenization analysis for the problem with separated nonlinear mechanisms. The decomposition of the nonlinearity into distinct components $\eta\phi(u_\varepsilon)$ and $\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon)$ necessitates a refined analytical approach that can handle their different scaling behaviors and interaction with the multiple temporal scales. These foundations are essential for the rigorous formulation of the global homogenized problem and provide the technical machinery required for the proof of our main convergence theorem.

3.1. Functional Analytic Setting for Separated Nonlinearities

We begin by recalling the identification of the space $L^2_{\text{per}}(Y)$ with $L^2(\mathbb{T}^N)$, where $\mathbb{T}^N$ denotes the N -dimensional torus. This identification allows us to recognize $L^2_{\text{per}}(Y)$ as a Hilbert space that is self-dual under the standard L^2 inner product. Within this framework, we consider the Gelfand triple (evolution triple):

$$W_{\#}^{1,2}(Y) \hookrightarrow L_{\#}^2(Y) \hookrightarrow (W_{\#}^{1,2}(Y))',$$

where both embeddings are continuous and dense, and $(W_{\#}^{1,2}(Y))'$ denotes the topological dual of $W_{\#}^{1,2}(Y)$.

If $[\cdot, \cdot]$ represents the duality pairing between $(W_{\#}^{1,2}(Y))'$ and $W_{\#}^{1,2}(Y)$, then for every $u \in L_{\#}^2(Y)$ and $\varphi \in W_{\#}^{1,2}(Y)$, we have

$$[u, \varphi] = \int_Y u(y) \varphi(y) \, dy.$$

The space $W_{\#}^{1,2}(Y)$, equipped with the norm

$$\|v\|_{\#} = \left(\sum_{i=1}^N \left\| \frac{\partial v}{\partial y_i} \right\|_{L^2(Y)}^2 \right)^{1/2}, \quad v \in W_{\#}^{1,2}(Y),$$

forms a Hilbert space. Moreover, by standard duality theory, the space $L_{\text{per}}^2(Z^2; (W_{\#}^{1,2}(Y))')$ is isometrically isomorphic to the dual of $L_{\text{per}}^2(Z^2; W_{\#}^{1,2}(Y))$.

To accommodate the separated nonlinear structure, we introduce the product space

$$\mathcal{H} = L_{\text{per}}^2(Z^2; L^{\infty}(Y)) \times L^2(Q \times Y \times Z^2),$$

which will capture the limiting behavior of the nonlinear terms $\eta\phi(u_{\varepsilon})$ and $\psi^{\varepsilon}(\cdot, \cdot, \nabla u_{\varepsilon})$ in the multiscale limit. Note that the second component is L^2 due to the linear growth of ψ in its gradient argument.

3.2. Test Function Spaces and Density Results

We now introduce the fundamental space of test functions adapted to our model problem. Following the standard framework for evolution multiscale convergence [2], we define

$$\mathcal{E} = \mathcal{C}_{\text{per}}^{\infty}(Z^2) \otimes \mathcal{C}_{\#}^{\infty}(Y),$$

where $\mathcal{C}_{\text{per}}^{\infty}(Z^2)$ denotes smooth functions periodic in both τ and ζ , and $\mathcal{C}_{\#}^{\infty}(Y)$ denotes smooth periodic functions with zero mean in Y . This space is dense in the spaces required for our homogenization analysis.

Lemma 1. *Let $u \in \mathcal{D}'_{\text{per}}(Y \times Z^2)$. Consider the restriction $u|_{\mathcal{E}}$ of u to $\mathcal{E}$. If u is continuous on $\mathcal{E}$ with respect to the norm topology of $L_{\text{per}}^2(Z^2; W_{\#}^{1,2}(Y))$, then $u \in L_{\text{per}}^2(Z^2; (W_{\#}^{1,2}(Y))')$ and we have*

$$\langle u, \varphi \rangle = \int_0^1 \int_0^1 [u(\tau, \zeta), \varphi(\cdot, \tau, \zeta)] \, d\tau \, d\zeta$$

for every $\varphi \in \mathcal{E}$, where $\langle \cdot, \cdot \rangle$ denotes the duality pairing between $\mathcal{D}'_{\text{per}}(Y \times Z^2)$ and $\mathcal{D}_{\text{per}}(Y \times Z^2)$, and the right-hand side denotes the duality pairing between $L_{\text{per}}^2(Z^2; (W_{\#}^{1,2}(Y))')$ and $L_{\text{per}}^2(Z^2; W_{\#}^{1,2}(Y))$.

Proof. The continuity assumption implies that u extends uniquely to a continuous linear functional on $L_{\text{per}}^2(Z^2; W_{\#}^{1,2}(Y))$. By the Riesz representation theorem for Bochner spaces, this functional is represented by an element of

$L^2_{\text{per}}(Z^2; (W^{1,2}_{\#}(Y))')$, and the duality formula follows by standard approximation arguments. $\square$

3.3. The Microscopic Resolution Space for Nonlinearities

We define the central function space for our microscopic analysis:

$$\mathcal{V} := \left\{ v = v(y, \tau, \zeta) \in L^2_{\text{per}}(Z^2; W^{1,2}_{\#}(Y)) : \frac{\partial^2 v}{\partial \tau^2} \in L^2_{\text{per}}(Z^2; (W^{1,2}_{\#}(Y))') \right\},$$

where the derivative $\frac{\partial^2 v}{\partial \tau^2}$ is understood in the distributional sense within $\mathcal{D}'_{\text{per}}(Y \times Z^2)$. The space $\mathcal{V}$ forms a reflexive Banach space when equipped with the norm

$$\|v\|_{\mathcal{V}} = \left(\|v\|_{L^2_{\text{per}}(Z^2; W^{1,2}_{\#}(Y))}^2 + \left\| \frac{\partial^2 v}{\partial \tau^2} \right\|_{L^2_{\text{per}}(Z^2; (W^{1,2}_{\#}(Y))')}^2 \right)^{\frac{1}{2}}.$$

This space captures the essential regularity requirements for the corrector functions that will appear in the homogenization process. The ζ -scale (the faster timescale) will be treated through averaging, and does not require derivative regularity in the corrector space.

Lemma 2. *The space $\mathcal{E}$ is dense in $\mathcal{V}$ with respect to the $\mathcal{V}$ -norm topology.*

Proof. This is a standard result in evolution multiscale convergence theory; see [2, 11]. The density follows from the fact that smooth periodic functions in (τ, ζ, y) are dense in the spaces $L^2_{\text{per}}(Z^2; W^{1,2}_{\#}(Y))$ and $L^2_{\text{per}}(Z^2; (W^{1,2}_{\#}(Y))')$. The result is obtained by a standard regularization and truncation argument. $\square$

The density result established in Lemma 2 is crucial for our homogenization methodology, as it allows us to work with smooth test functions when establishing the limiting equations for both nonlinear components, then extend the results to the entire space $\mathcal{V}$ by density arguments.

Corollary 3.1. *Let $(u_{\varepsilon})_{\varepsilon \in E}$ be as in Proposition 2.4. There exist a subsequence $E' \subset E$ and a pair $(u_0, u_1) \in \mathcal{W} \times L^2(Q; \mathcal{V})$ such that the following convergences hold:*

- $u_{\varepsilon} \rightharpoonup u_0$ in $\mathcal{W}$ -weak,
- $\nabla u_{\varepsilon} \xrightarrow{\text{m.s.}} \nabla u_0 + \nabla_y u_1$ in $L^2(Q)$ -weak,
- $\eta^{\varepsilon} \phi(u_{\varepsilon}) \xrightarrow{\text{m.s.}} \eta(y, \tau, \zeta) \phi(u_0)$ in $L^2(Q)$ -weak.

Moreover, the following preliminary convergence holds for the gradient-dependent nonlinearity:

$$\psi^{\varepsilon}(\cdot, \cdot, \nabla u_{\varepsilon}) \xrightarrow{\text{m.s.}} \Psi(x, t, y, \tau, \zeta) \quad \text{in } L^2(Q)\text{-weak},$$

for some $\Psi \in L^2(Q \times Y \times Z^2)$. The identification

$$\Psi(x, t, y, \tau, \zeta) = \psi(y, \tau, \zeta, \nabla u_0 + \nabla_y u_1)$$

will be established in Sect. 4 using a compensated compactness argument that exploits the Lipschitz continuity of ψ and the strong convergence of the gra-

dients obtained from the energy dissipation structure. The weak convergence stated here is sufficient for the formal derivation of the cell problems [20, 21].

Proof. The result follows by combining Proposition 2.4 with the density of $\mathcal{E}$ in $\mathcal{V}$ established in Lemma 2. The convergence of the nonlinear terms follows from the structural assumptions (H3) and the strong convergence properties established in Proposition 2.4. The conclusion then follows from the abstract framework for multiscale convergence developed in [2], adapted to account for the separated nonlinear structure. $\square$

3.4. Rescaled Solution Convergence

We conclude this section with a crucial convergence result for the rescaled solution sequence, which plays a fundamental role in identifying the corrector function in the homogenization process.

Proposition 3.1. *Let u_ε be the solution of (1.1) and define the rescaled function*

$$v_\varepsilon(x, t, y, \tau, \zeta) := u_\varepsilon(x, t) \quad \text{for } (x, t) \in Q, (y, \tau, \zeta) \in Y \times Z^2.$$

This definition means that v_ε is the extension of u_ε to the microscopic variables (y, τ, ζ) , constant in those variables. Then there exists $v_0 \in L^2(Q; \mathcal{V})$ such that, up to a subsequence,

$$v_\varepsilon \rightharpoonup v_0 \quad \text{in } L^2(Q \times Y \times Z^2)\text{-weak},$$

and the following identification holds:

$$v_0(x, t, y, \tau, \zeta) = u_0(x, t) + u_1(x, t, y, \tau, \zeta),$$

where u_0 is the weak limit of u_ε in $\mathcal{W}$ and $u_1 \in L^2(Q; \mathcal{V})$ is the corrector function.

Proof. The proof follows from the uniform boundedness of u_ε in $\mathcal{W}$ provided by Proposition 2.1 and the compactness properties of the embedding $\mathcal{W} \hookrightarrow L^2(Q)$. The identification of the limit follows from the gradient decomposition in Proposition 2.4 and the structure of the space $\mathcal{V}$. The periodicity in the microscopic variables is inherited from the periodicity assumptions in (H1). $\square$

Proposition 3.2. *Let $(u_\varepsilon)_{\varepsilon \in E}$, E' , and $\mathbf{u} = (u_0, u_1)$ be as in Corollary 3.1. Then, for all $\varphi \in \mathcal{D}(Q) \otimes \mathcal{E}$ with φ having zero mean in Y , we have*

$$\int_Q \frac{1}{\varepsilon} u_\varepsilon \varphi^\varepsilon \, dx \, dt \longrightarrow \iiint_{Q \times Y \times Z^2} u_1 \varphi \, dx \, dt \, dy \, d\tau \, d\zeta \quad \text{as } E' \ni \varepsilon \rightarrow 0, \quad (3.1)$$

where the oscillating test function $\varphi^\varepsilon \in \mathcal{D}(Q)$ is defined by

$$\varphi^\varepsilon(x, t) = \varphi\left(x, t, \frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}\right) \quad \text{for } (x, t) \in Q.$$

Proof. Let $\varphi \in \mathcal{D}(Q) \otimes \mathcal{E}$ with zero mean in Y . Since φ has zero mean in the microscopic spatial variable, i.e.,

$$\int_Y \varphi(\cdot, \cdot, y, \cdot, \cdot) \, dy = 0$$

for almost every (x, t, τ, ζ) , the Fredholm alternative for the periodic Laplacian guarantees the existence of $\theta \in \mathcal{D}(Q) \otimes \mathcal{E}$ such that

$$\varphi = \Delta_y \theta.$$

For $\varepsilon > 0$, set

$$\theta^\varepsilon(x, t) := \theta \left(x, t, \frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2} \right).$$

From the chain rule, we have the identity

$$\frac{1}{\varepsilon}(\Delta_y \theta)^\varepsilon = \operatorname{div}_x((\nabla_y \theta)^\varepsilon) - (\operatorname{div}_x \nabla_y \theta)^\varepsilon,$$

where div_x denotes the divergence with respect to the slow variable $x \in Q$. Therefore, integrating over Q and using integration by parts (the boundary terms vanish since φ and hence θ have compact support in Q) yields

$$\begin{aligned} \int_Q \frac{1}{\varepsilon} u_\varepsilon \varphi^\varepsilon \, dx \, dt &= \int_Q \frac{1}{\varepsilon} u_\varepsilon (\Delta_y \theta)^\varepsilon \, dx \, dt \\ &= \int_Q u_\varepsilon \operatorname{div}_x((\nabla_y \theta)^\varepsilon) \, dx \, dt - \int_Q u_\varepsilon (\operatorname{div}_x \nabla_y \theta)^\varepsilon \, dx \, dt \\ &= - \int_Q \nabla_x u_\varepsilon \cdot (\nabla_y \theta)^\varepsilon \, dx \, dt - \int_Q u_\varepsilon (\operatorname{div}_x \nabla_y \theta)^\varepsilon \, dx \, dt. \end{aligned}$$

We now pass to the limit along $E' \ni \varepsilon \rightarrow 0$. By Corollary 3.1, u_0 is independent of y . Moreover, since θ is periodic in y , we have

$$\int_Y \nabla_y \theta \, dy = 0.$$

Using Proposition 2.4 and the two-scale convergence of $\nabla_x u_\varepsilon$ to $\nabla_x u_0 + \nabla_y u_1$, we obtain

$$\int_Q \nabla_x u_\varepsilon \cdot (\nabla_y \theta)^\varepsilon \, dx \, dt \longrightarrow \iiint_{Q \times Y \times Z^2} (\nabla_x u_0 + \nabla_y u_1) \cdot \nabla_y \theta \, dx \, dt \, dy \, d\tau \, d\zeta.$$

Since u_0 is independent of y , the contribution involving $\nabla_x u_0$ vanishes:

$$\iiint_{Q \times Y \times Z^2} \nabla_x u_0 \cdot \nabla_y \theta = \int_Q \nabla_x u_0 \cdot \left(\int_Y \nabla_y \theta \, dy \right) = 0.$$

Likewise, for the remaining term,

$$\begin{aligned} \int_Q u_\varepsilon (\operatorname{div}_x \nabla_y \theta)^\varepsilon \, dx \, dt &\longrightarrow \iiint_{Q \times Y \times Z^2} u_0 \operatorname{div}_x \nabla_y \theta \, dx \, dt \, dy \, d\tau \, d\zeta \\ &= \int_Q u_0 \operatorname{div}_x \left(\int_Y \nabla_y \theta \, dy \right) = 0. \end{aligned}$$

Hence,

$$\int_Q \frac{1}{\varepsilon} u_\varepsilon \varphi^\varepsilon \, dx \, dt \rightarrow - \iiint_{Q \times Y \times Z^2} \nabla_y u_1 \cdot \nabla_y \theta \, dx \, dt \, dy \, d\tau \, d\zeta.$$

Finally, using the identity

$$\begin{aligned} - \iiint_{Q \times Y \times Z^2} \nabla_y u_1 \cdot \nabla_y \theta \, dx \, dt \, dy \, d\tau \, d\zeta &= \iiint_{Q \times Y \times Z^2} u_1 \Delta_y \theta \, dx \, dt \, dy \, d\tau \, d\zeta \\ &= \iiint_{Q \times Y \times Z^2} u_1 \varphi \, dx \, dt \, dy \, d\tau \, d\zeta, \end{aligned}$$

we conclude that

$$\int_Q \frac{1}{\varepsilon} u_\varepsilon \varphi^\varepsilon \, dx \, dt \rightarrow \iiint_{Q \times Y \times Z^2} u_1 \varphi \, dx \, dt \, dy \, d\tau \, d\zeta.$$

This proves the desired convergence (3.1). $\square$

4. Homogenization Results

4.1. Convergence Results for Separated Nonlinearities

Before presenting and proving the main homogenization theorem, we establish several crucial convergence results that form the technical foundation of our analysis for the problem with separated nonlinear mechanisms.

Proposition 4.1. *Let $(u_\varepsilon)_{\varepsilon \in E}$ be a sequence in $L^p(Q)$, where E is a fundamental sequence, such that $u_\varepsilon \rightarrow u_0$ strongly in $L^p(Q)$ as $\varepsilon \rightarrow 0$ along E , with $u_0 \in L^p(Q)$ and $p = \gamma + 2$, where γ is defined in assumption **(H3)**. Then, as $\varepsilon \rightarrow 0$ in E , we have*

$$\eta^\varepsilon \phi(u_\varepsilon) \xrightarrow{\text{m.s.}} \eta(y, \tau, \zeta) \phi(u_0) \quad \text{in } L^2(Q)\text{-weak},$$

where

$$\eta^\varepsilon(x, t) := \eta\left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon}, \frac{t}{\varepsilon^2}\right).$$

Proof. First, observe that the regularity assumptions on η and ϕ ensure that $\eta(y, \tau, \zeta) \phi(u_0) \in L^2(Q; \mathcal{C}_{\text{per}}(Y \times Z^2))$. Therefore, by the properties of multi-scale convergence,

$$\eta^\varepsilon \phi(u_0) \xrightarrow{\text{m.s.}} \eta(y, \tau, \zeta) \phi(u_0) \quad \text{in } L^2(Q)\text{-weak}.$$

To complete the proof, it suffices to demonstrate that

$$\eta^\varepsilon [\phi(u_\varepsilon) - \phi(u_0)] \xrightarrow{\text{m.s.}} 0 \quad \text{in } L^2(Q)\text{-weak}.$$

Define the difference term

$$a_\varepsilon := \eta^\varepsilon [\phi(u_\varepsilon) - \phi(u_0)].$$

From conditions (i) and (iii) of assumption **(H3)** and the strong convergence $u_\varepsilon \rightarrow u_0$ in $L^p(Q)$, it follows that the sequence $(a_\varepsilon)_{\varepsilon \in E}$ is bounded in $L^2(Q)$. Consequently, there exists a subsequence $E' \subset E$ and a function $\chi \in L^2(Q \times Y \times Z^2)$ such that

$$a_\varepsilon \xrightarrow{\text{m.s.}} \chi \quad \text{in } L^2(Q)\text{-weak as } \varepsilon \rightarrow 0 \text{ in } E'. \quad (4.1)$$

Now, for any test function $g \in L^2(Q; \mathcal{C}_{\text{per}}(Y \times Z^2))$, the mean value theorem and condition (iii) of **(H3)** yield

$$\begin{aligned} \left| \int_Q a_\varepsilon g^\varepsilon dx dt \right| &\leq \|\eta\|_{L^\infty} \int_Q |\phi(u_\varepsilon) - \phi(u_0)| |g^\varepsilon| dx dt \\ &\leq \|\eta\|_{L^\infty} c_4 \int_Q (1 + |u_\varepsilon|^\gamma + |u_0|^\gamma) |u_\varepsilon - u_0| |g^\varepsilon| dx dt \\ &\leq C \left(\|u_\varepsilon\|_{L^p(Q)}^\gamma + \|u_0\|_{L^p(Q)}^\gamma + 1 \right) \|u_\varepsilon - u_0\|_{L^p(Q)} \|g^\varepsilon\|_{L^2(Q)}, \end{aligned}$$

where the last inequality follows from Holder's inequality with exponents satisfying

$$\frac{\gamma}{2(\gamma+1)} + \frac{1}{2(\gamma+1)} + \frac{1}{2} = 1.$$

Passing to the limit as $\varepsilon \rightarrow 0$ in E' and applying (4.1), we obtain

$$\iiint_{Q \times Y \times Z^2} \chi g dx dt dy d\tau d\zeta = 0.$$

Since g was arbitrary in $L^2(Q; \mathcal{C}_{\text{per}}(Y \times Z^2))$, it follows that $\chi = 0$ almost everywhere. Therefore, every multiscale limit of $(a_\varepsilon)_{\varepsilon \in E}$ vanishes, and the entire sequence converges multiscale-weakly to zero, which completes the proof. $\square$

4.2. Variational Formulation

Let us first define the space of test functions adapted to our problem:

$$F_0^\infty = \mathcal{D}(Q) \times (\mathcal{D}(Q) \otimes \mathcal{E}).$$

Since $\mathcal{E}$ is dense in $\mathcal{V}$, the space F_0^∞ is dense in $\mathbb{F}_0^1 := \mathcal{W}_0 \times L^2(Q; \mathcal{V})$.

We introduce the bilinear form a_Q defined by

$$a_Q(\mathbf{u}, \mathbf{v}) = \sum_{i,j=1}^N \iiint_{Q \times Y \times Z^2} a_{ij}(y) \mathbb{D}_j \mathbf{u} \mathbb{D}_i \mathbf{v} dx dt dy d\tau d\zeta \quad (\mathbf{u}, \mathbf{v} \in \mathbb{F}_0^1),$$

where for $\mathbf{w} = (w_0, w_1) \in \mathbb{F}_0^1$, we define the combined gradient

$$\mathbb{D}_j \mathbf{w} = \frac{\partial w_0}{\partial x_j} + \frac{\partial w_1}{\partial y_j}.$$

Consider the following homogenized variational problem for our system:

Find $\mathbf{u} = (u_0, u_1) \in \mathbb{F}_0^1$ such that for all $\mathbf{v} = (v_0, v_1) \in \mathbb{F}_0^1$,

$$\begin{aligned} \int_0^T \langle u_0''(t), v_0(t) \rangle dt + \int_Q \left(\int_0^1 \int_0^1 \left[\frac{\partial^2 u_1}{\partial \tau^2}(\tau, \zeta), v_1(\tau, \zeta) \right] d\tau d\zeta \right) dx dt \\ + a_Q(\mathbf{u}, \mathbf{v}) + \int_Q [\tilde{\eta} \phi(u_0) + \tilde{\psi}(\mathbb{D} \mathbf{u})] v_0 dx dt = \int_Q f v_0 dx dt. \end{aligned} \quad (4.2)$$

Here, the averaged nonlinearities are defined by

$$\begin{aligned}\widetilde{\eta\phi}(u_0) &= \iiint_{Y \times Z^2} \eta(y, \tau, \zeta) \phi(u_0) \, dy \, d\tau \, d\zeta, \quad \widetilde{\psi}(\mathbb{D}\mathbf{u}) \\ &= \iiint_{Y \times Z^2} \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u}) \, dy \, d\tau \, d\zeta.\end{aligned}$$

Lemma 3. *The homogenized variational problem (4.2) has at most one solution.*

Proof. We suppose that $\mathbf{u} = (u_0, u_1)$ and $\widetilde{\mathbf{u}} = (\widetilde{u}_0, \widetilde{u}_1)$ are two solutions of the system (4.2) with the same initial data and source term. Define the differences:

$$w_0 = u_0 - \widetilde{u}_0, \quad w_1 = u_1 - \widetilde{u}_1.$$

We have

$$w_0(0) = \frac{\partial w_0}{\partial t}(0) = 0 \text{ and let us set } \mathbf{w} = \mathbf{u} - \widetilde{\mathbf{u}} = (w_0, w_1) \in \mathbb{F}_0^1.$$

Subtracting the variational formulations satisfied by each solution, we obtain the following coupled system

$$\begin{aligned}\int_0^T \langle w_0''(t), v_0(t) \rangle \, dt + \int_Q \left(\int_0^1 \int_0^1 \left[\frac{\partial^2 w_1}{\partial \tau^2}(\tau, \zeta), v_1(\tau, \zeta) \right] \, d\tau \, d\zeta \right) \, dx \, dt \\ + a_Q(\mathbf{w}, \mathbf{v}) + \int_Q \left[\widetilde{\eta\phi}(u_0) - \widetilde{\eta\phi}(\widetilde{u}_0) \right] v_0 \, dx \, dt \\ + \int_Q \left[\widetilde{\psi}(\mathbb{D}\mathbf{u}) - \widetilde{\psi}(\mathbb{D}\widetilde{\mathbf{u}}) \right] v_0 \, dx \, dt = 0.\end{aligned}\quad (4.3)$$

In (4.3) by choosing $\mathbf{v} = \mathbf{w}'$, we have

$$\begin{aligned}\int_0^T \left\langle \frac{\partial^2 w_0}{\partial t^2}(t), \frac{\partial w_0}{\partial t}(t) \right\rangle \, dt + \int_Q \left(\int_0^1 \int_0^1 \left[\frac{\partial^2 w_1}{\partial \tau^2}(\tau, \zeta), \frac{\partial w_1}{\partial \tau}(\tau, \zeta) \right] \, d\tau \, d\zeta \right) \, dx \, dt \\ + a_Q(\mathbf{w}, \mathbf{w}') + \int_Q \left[\widetilde{\eta\phi}(u_0) - \widetilde{\eta\phi}(\widetilde{u}_0) \right] \frac{\partial w_0}{\partial t} \, dx \, dt \\ + \int_Q \left[\widetilde{\psi}(\mathbb{D}\mathbf{u}) - \widetilde{\psi}(\mathbb{D}\widetilde{\mathbf{u}}) \right] \frac{\partial w_0}{\partial t} \, dx \, dt = 0.\end{aligned}\quad (4.4)$$

We are now going to evaluate each term of (4.4).

Concerning the first term, we have

$$\begin{aligned}\int_0^T \left\langle \frac{\partial^2 w_0}{\partial t^2}(t), \frac{\partial w_0}{\partial t}(t) \right\rangle \, dt &= \frac{1}{2} \left\| \frac{\partial w_0}{\partial t}(T) \right\|_{L^2(\Omega)}^2 - \frac{1}{2} \left\| \frac{\partial w_0}{\partial t}(0) \right\|_{L^2(\Omega)}^2 \\ &= \frac{1}{2} \left\| \frac{\partial w_0}{\partial t}(T) \right\|_{L^2(\Omega)}^2.\end{aligned}$$

We have

$$\begin{aligned}
 & \int_Q \left(\int_0^1 \int_0^1 \left[\frac{\partial^2 w_1}{\partial \tau^2}(\tau, \zeta), \frac{\partial w_1}{\partial \tau}(\tau, \zeta) \right] d\tau d\zeta \right) dx dt \\
 &= \int_Q \left(\int_0^1 \int_0^1 \frac{1}{2} \frac{d}{d\tau} \left\| \frac{\partial w_1}{\partial \tau} \right\|^2 d\tau d\zeta \right) dx dt \\
 &= \frac{1}{2} \int_Q \left(\int_0^1 \left[\left\| \frac{\partial w_1}{\partial \tau} \right\|^2 \right]_{\tau=0}^{\tau=1} d\zeta \right) dx dt \\
 &= 0,
 \end{aligned}$$

since w_1 is periodic with period 1 in τ .

Using the condition **(H2)**, we obtain

$$\begin{aligned}
 \int_0^T a_Q(\mathbf{w}, \mathbf{w}') dt &= \int_0^T \frac{1}{2} \frac{d}{dt} (a_Q(\mathbf{w}, \mathbf{w})) dt \\
 &= \frac{1}{2} a_Q(\mathbf{w}(T), \mathbf{w}(T)) \geq \frac{\alpha}{2} \|\mathbb{D}\mathbf{w}(T)\|_{L^2(\Omega)}^2.
 \end{aligned}$$

We now estimate the nonlinear terms.

Since $\eta > 0$ and ϕ is nondecreasing, the function $\widetilde{\eta\phi}$ is nondecreasing. Consequently,

$$(\widetilde{\eta\phi}(u_0) - \widetilde{\eta\phi}(\tilde{u}_0)) w_0 \geq 0 \quad \text{a.e. in } Q.$$

Define

$$\Phi(t) := \int_Q [\widetilde{\eta\phi}(u_0) - \widetilde{\eta\phi}(\tilde{u}_0)] w_0 dx.$$

Then $\Phi(t) \geq 0$ for all t , and

$$\frac{d}{dt} \Phi(t) = \int_Q [\widetilde{\eta\phi}(u_0) - \widetilde{\eta\phi}(\tilde{u}_0)] \frac{\partial w_0}{\partial t} dx.$$

Using the Lipschitz condition **(H3)**(vii) and Young's inequality,

$$\begin{aligned}
 & \left| \int_Q [\widetilde{\psi}(\mathbb{D}\mathbf{u}) - \widetilde{\psi}(\mathbb{D}\tilde{\mathbf{u}})] \frac{\partial w_0}{\partial t} dx \right| \leq c_7 \int_Q |\mathbb{D}w| \left| \frac{\partial w_0}{\partial t} \right| dx \\
 & \leq C \left(\left\| \frac{\partial w_0}{\partial t} \right\|_{L^2(\Omega)}^2 + \|\mathbb{D}w\|_{L^2(\Omega)}^2 \right).
 \end{aligned}$$

Integrating (4.4) from 0 to t and using $\Phi(t) - \Phi(0) \geq 0$ (since $\Phi(0) = 0$), and combining the estimates yield

$$\left\| \frac{\partial w_0}{\partial t}(t) \right\|_{L^2(\Omega)}^2 + \|\mathbb{D}w(t)\|_{L^2(\Omega)}^2 \leq C \int_0^t \left(\left\| \frac{\partial w_0}{\partial s}(s) \right\|_{L^2(\Omega)}^2 + \|\mathbb{D}w(s)\|_{L^2(\Omega)}^2 \right) ds.$$

Choosing

$$E(t) = \left\| \frac{\partial w_0}{\partial t}(t) \right\|_{L^2(\Omega)}^2 + \|\mathbb{D}w(t)\|_{L^2(\Omega)}^2,$$

then Gronwall's inequality gives $E(t) \equiv 0$, hence $\frac{\partial w_0}{\partial t}(t) \equiv 0$ and $\mathbb{D}w(t) = 0$. Since $w_0(0) \equiv 0$, we have $w_0(t) \equiv 0$ for all $t \in [0, T]$, from $\mathbb{D}w(t) = 0$ we conclude that $w_1 \equiv 0$. This proves the uniqueness. $\square$

We now define the reduced microscopic spaces:

$$\mathcal{E}_1 = \mathcal{D}_{\text{per}}(Z) \otimes \mathcal{D}_{\#}(Y),$$

$$\mathcal{V}_1 = \left\{ v \in L^2_{\text{per}}(Z; W_{\#}^{1,2}(Y)) : \frac{\partial^2 v}{\partial \tau^2} \in L^2_{\text{per}}\left(Z; (W_{\#}^{1,2}(Y))'\right) \right\}.$$

The space $\mathcal{E}_1$ is dense in $\mathcal{V}_1$, and by reasoning analogous to Lemma 3, the associated variational problem in the space $\mathbb{G}_0^1 = \mathcal{W}_0 \times L^2(Q; \mathcal{V}_1)$ admits at most one solution. This reduced framework will be employed in the proof of our main homogenization theorem.

4.3. Main Homogenization Theorem

We now state and prove the main homogenization theorem for the nonlinear hyperbolic equation with separated nonlinearities and multiple temporal scales.

Theorem 4.1. *Let u_ε be the unique solution of the nonlinear hyperbolic problem (1.1) under assumptions (H1)–(H3). Then, as the scaling parameter $\varepsilon \rightarrow 0$, the following convergence results hold:*

- $u_\varepsilon \rightharpoonup u_0$ weakly in the energy space $\mathcal{W}$,
- $\nabla u_\varepsilon \xrightarrow{\text{m.s.}} \nabla u_0 + \nabla_y u_1$ in $L^2(Q)$ -weak multiscale sense,
- $\eta^\varepsilon \phi(u_\varepsilon) \xrightarrow{\text{m.s.}} \eta(y, \tau, \zeta) \phi(u_0)$ in $L^2(Q)$ -weak,
- $\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \xrightarrow{\text{m.s.}} \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u})$ in $L^2(Q)$ -weak,

where $\mathbf{u} = (u_0, u_1) \in \mathbb{F}_0^1$ denotes the unique solution pair of the homogenized variational problem (4.2), with $\mathbb{D}\mathbf{u} = \nabla u_0 + \nabla_y u_1$ representing the combined macroscopic-microscopic gradient.

Proof. The homogenization result is established through a synthesis of compactness arguments and multiscale convergence techniques. Beginning with the uniform boundedness of the sequence (u_ε) in $\mathcal{W}$ provided by Theorem 2.1, we apply the compactness results together with the gradient decomposition from Proposition 2.4. This yields a subsequence (retaining the notation ε) and a limit pair $\mathbf{u} = (u_0, u_1) \in \mathbb{F}_0^1$ such that $u_\varepsilon \rightharpoonup u_0$ weakly in $\mathcal{W}$ and $\nabla u_\varepsilon \xrightarrow{\text{m.s.}} \nabla u_0 + \nabla_y u_1$ in the $L^2(Q)$ -weak multiscale sense.

The convergence of the nonlinear amplitude-dependent term follows from Proposition 4.1, leveraging the strong $L^p(Q)$ convergence of u_ε which stems from the compact embedding of $\mathcal{W}$.

We now prove the strong convergence of the gradients, which is essential for identifying the limit of the gradient-dependent nonlinearity. From the energy estimates in Proposition 2.1, we have the additional bound

$$\int_Q |\partial_t u_\varepsilon|^p \, dx \, dt \leq C,$$

with $p = \gamma + 2$. By interpolation and the uniform bound in $L^\infty(0, T; H_0^1(\Omega))$, this implies that $\partial_t u_\varepsilon$ is strongly compact in $L^2(0, T; L^2(\Omega))$ by the Aubin–Lions lemma. Specifically, we have the uniform bound

$$\|\partial_t u_\varepsilon\|_{L^\infty(0, T; L^2(\Omega))} + \|\partial_t u_\varepsilon\|_{L^p(Q)} \leq C,$$

which, together with the bound on $\partial_t^2 u_\varepsilon$ in $L^2(0, T; H^{-1}(\Omega))$, yields strong convergence of $\partial_t u_\varepsilon$ in $L^2(Q)$.

From the equation structure,

$$\nabla \cdot (A^\varepsilon \nabla u_\varepsilon) = \partial_t^2 u_\varepsilon + \eta^\varepsilon \phi(u_\varepsilon) + \psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) - f,$$

and using the uniform estimates for the terms on the right-hand side (from Proposition 2.1), we obtain uniform bounds on $\nabla \cdot (A^\varepsilon \nabla u_\varepsilon)$ in $L^2(0, T; H^{-1}(\Omega))$. Together with the bound on u_ε in $L^\infty(0, T; H_0^1(\Omega))$, the Aubin–Lions lemma yields strong compactness of u_ε in $L^2(0, T; H_0^1(\Omega))$. Hence, ∇u_ε converges strongly in $L^2(Q)$:

$$\nabla u_\varepsilon \xrightarrow{\text{m.s.}} \nabla u_0 + \nabla_y u_1 \quad \text{strongly in } L^2(Q).$$

With the strong convergence of the gradients established, we now identify the limit of the gradient-dependent nonlinearity. Since ψ is Lipschitz in its gradient argument **(H3)**(vii) and of linear growth **(H3)**(v), we have

$$\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \xrightarrow{\text{m.s.}} \psi(y, \tau, \zeta, \nabla u_0 + \nabla_y u_1) \quad \text{in } L^2(Q).$$

Indeed, this follows from the dominated convergence theorem: the strong convergence of ∇u_ε to $\mathbb{D}\mathbf{u}$ in $L^2(Q)$ implies, by the Lipschitz property, that $\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) \rightarrow \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u})$ strongly in $L^2(Q)$, since

$$\|\psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) - \psi(y, \tau, \zeta, \mathbb{D}\mathbf{u})\|_{L^2(Q)} \leq c_7 \|\nabla u_\varepsilon - (\mathbb{D}\mathbf{u})^\varepsilon\|_{L^2(Q)} \rightarrow 0.$$

This proves the fourth convergence item in the theorem.

To pass to the limit in the variational formulation of the original problem (1.1), we consider test functions $\mathbf{v} = (v_0, v_1) \in F_0^\infty$. Multiplying the original equation by $v_0 + \varepsilon v_1^\varepsilon$ and integrating by parts, we obtain the weak formulation:

$$\begin{aligned} & \int_0^T \langle u_\varepsilon''(t), v_0(t) \rangle dt + \int_Q A^\varepsilon \nabla u_\varepsilon \cdot \nabla v_0 \, dx dt + \int_Q \eta^\varepsilon \phi(u_\varepsilon) v_0 \, dx dt \\ & + \int_Q \psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) v_0 \, dx dt + \int_Q A^\varepsilon \nabla u_\varepsilon \cdot \nabla v_1^\varepsilon \, dx dt \\ & + \varepsilon \int_Q \eta^\varepsilon \phi(u_\varepsilon) v_1^\varepsilon \, dx dt + \varepsilon \int_Q \psi^\varepsilon(\cdot, \cdot, \nabla u_\varepsilon) v_1^\varepsilon \, dx dt \\ & = \int_Q f v_0 \, dx dt + \varepsilon \int_Q f v_1^\varepsilon \, dx dt. \end{aligned}$$

Using the convergence results above and passing to the limit $\varepsilon \rightarrow 0$, the terms involving $\varepsilon v_1^\varepsilon$ vanish due to the factor of ε . The remaining terms yield exactly the homogenized variational problem (4.2). The key step is the identification of the limit of the $A^\varepsilon \nabla u_\varepsilon \cdot \nabla v_1^\varepsilon$ term, which produces the bilinear form $a_Q(\mathbf{u}, (0, v_1))$, and the $A^\varepsilon \nabla u_\varepsilon \cdot \nabla v_0$ term, which produces $a_Q(\mathbf{u}, (v_0, 0))$.

Finally, the uniqueness result in Lemma 3 guarantees that the homogenized variational problem possesses at most one solution. This uniqueness, combined with the compactness of the sequence, implies the convergence of the entire sequence (u_ε) , not merely a subsequence, thereby completing the proof of the homogenization theorem. $\square$

Remark 4.1. The homogenized system exhibits several notable features:

- The macroscopic equation is coupled to a microscopic wave equation in the fast time variable τ .
- The separated nonlinearities yield distinct averaged contributions $\widetilde{\eta\phi}(u_0)$ and $\widetilde{\psi}(\mathbb{D}\mathbf{u})$.
- The multiple temporal scales induce nonlocal memory effects in the effective dissipation.
- The corrector u_1 captures the fine-scale oscillations and their interaction with the nonlinear mechanisms.

4.4. Effective Macroscopic Equation

We now derive the effective macroscopic equation from the coupled system (4.2). The derivation proceeds in two steps: first, we solve the microscopic equation for the corrector u_1 in terms of u_0 ; second, we substitute this expression into the macroscopic equation.

Step 1: The microscopic equation. The microscopic part of (4.2) is obtained by taking test functions $\mathbf{v} = (0, \varphi_1)$ with $\varphi_1 \in L^2(Q; \mathcal{V})$. This yields:

$$\int_Q \left(\int_0^1 \int_0^1 \left[\frac{\partial^2 u_1}{\partial \tau^2}(\tau, \zeta), \varphi_1(\tau, \zeta) \right] d\tau d\zeta \right) dx dt + a_Q(\mathbf{u}, (0, \varphi_1)) = 0.$$

Integrating by parts in space and using the arbitrariness of φ_1 , we obtain the strong form of the microscopic equation:

$$\frac{\partial^2 u_1}{\partial \tau^2} - \nabla_y \cdot (A(y) \nabla_y u_1) = \nabla_y \cdot (A(y) \nabla u_0) \quad \text{in } Y \times Z^2, \quad (4.5)$$

with periodic boundary conditions in y and Z^2 , and the normalization condition $\int_Y u_1 dy = 0$.

Remark 4.2. The microscopic equation (4.5) depends only on the fast time variable $\tau = t/\varepsilon$, not on the faster scale $\zeta = t/\varepsilon^2$. This is because the ζ -scale oscillations in the original equation are treated through averaging over the period Z in the variational formulation. The corrector u_1 therefore captures only the intermediate timescale oscillations, while the ζ -scale contributions are homogenized and appear in the averaged nonlinearity $\widetilde{\psi}$. This separation of scales is a key feature of the three-scale structure of our problem.

The linearity of (4.5) implies that u_1 depends linearly on ∇u_0 . Specifically, by the method of separation of variables, we seek a solution of the form

$$u_1(x, t, y, \tau, \zeta) = \sum_{k=1}^N \chi_k(y, \tau) \frac{\partial u_0}{\partial x_k}(x, t),$$

where χ_k satisfies the cell problem:

$$\begin{cases} \frac{\partial^2 \chi_k}{\partial \tau^2} - \nabla_y \cdot (A(y) \nabla_y \chi_k) = \sum_{i=1}^N \frac{\partial a_{ik}}{\partial y_i}(y) & \text{in } Y \times Z, \\ \chi_k \text{ is } Y\text{-periodic in } y, \quad Z\text{-periodic in } \tau, \\ \int_Y \chi_k(y, \tau) dy = 0 & \text{for a.e. } \tau. \end{cases} \quad (4.6)$$

Note that χ_k is independent of ζ ; the ζ -dependence is treated by averaging in the definition of the effective nonlinearity.

Remark 4.3. In the special case where the fast time dynamics are negligible (i.e., the ∂_τ^2 term is omitted), the cell problem reduces to the classical static corrector:

$$-\nabla_y \cdot (A(y) \nabla_y \chi_k) = \sum_{i=1}^N \frac{\partial a_{ik}}{\partial y_i}(y), \quad \chi_k \in W_{\#}^{1,2}(Y).$$

However, in our general framework, the full cell problem (4.6) must be used.

Step 2: The macroscopic equation. The macroscopic part of (4.2) is obtained by taking test functions $\mathbf{v} = (\varphi_0, 0)$ with $\varphi_0 \in \mathcal{W}_0$. This yields:

$$\begin{aligned} \int_0^T \langle u_0''(t), \varphi_0(t) \rangle dt + a_Q(\mathbf{u}, (\varphi_0, 0)) + \int_Q [\widetilde{\eta}\phi(u_0) + \widetilde{\psi}(\mathbb{D}\mathbf{u})] \varphi_0 dx dt \\ = \int_Q f \varphi_0 dx dt. \end{aligned}$$

Using the representation $u_1 = \sum_k \chi_k \partial_{x_k} u_0$ and integrating by parts, we obtain the strong form of the macroscopic equation:

$$\frac{\partial^2 u_0}{\partial t^2} - \nabla \cdot (A^{\text{hom}} \nabla u_0) + \widetilde{\eta}\phi(u_0) + \widetilde{\psi}^{\text{eff}}(\nabla u_0) = f \quad \text{in } Q,$$

where the homogenized matrix is given by

$$A_{ij}^{\text{hom}} = \int_Y \left(a_{ij}(y) + \sum_{k=1}^N a_{ik}(y) \frac{\partial \chi_j}{\partial y_k} \right) dy,$$

with χ_j solving the static version of (4.6) (i.e., neglecting the ∂_τ^2 term in the special case).

The effective dissipation operator is defined by

$$\widetilde{\psi}^{\text{eff}}(\nabla u_0)(x, t) := \iiint_{Y \times Z^2} \psi \left(y, \tau, \zeta, \nabla u_0(x, t) + \sum_{k=1}^N \nabla_y \chi_k(y, \tau) \frac{\partial u_0}{\partial x_k}(x, t) \right) dy d\tau d\zeta.$$

In general, since χ_k depends on τ and ζ , the operator $\widetilde{\psi}^{\text{eff}}$ is nonlocal in time, involving convolutions of ∇u_0 with the Green's function of the fast time wave operator.

Summary. Eliminating the corrector u_1 from the coupled system (4.2) yields the following effective macroscopic equation:

$$\begin{cases} \frac{\partial^2 u_0}{\partial t^2} - \nabla \cdot (A^{\text{hom}} \nabla u_0) + \widetilde{\eta} \phi(u_0) + \widetilde{\psi}^{\text{eff}}(\nabla u_0) = f & \text{in } Q, \\ u_0 = 0 & \text{on } \partial\Omega \times (0, T), \\ u_0(x, 0) = \frac{\partial u_0}{\partial t}(x, 0) = 0 & \text{in } \Omega, \end{cases} \quad (4.7)$$

with A^{hom} defined as above and $\widetilde{\psi}^{\text{eff}}$ given by the explicit formula above.

Remark 4.4. Several important observations are in order:

- (1) The homogenized matrix A^{hom} is the classical one from periodic homogenization, depending only on the spatial microstructure $A(y)$.
- (2) The effective nonlinearity $\widetilde{\psi}^{\text{eff}}$ is generally nonlocal in time because the corrector χ_k depends on the fast time variable τ . This nonlocality is a direct consequence of the intermediate temporal scale in the original problem.
- (3) The amplitude-dependent term $\widetilde{\eta} \phi(u_0)$ remains local in time because the ϕ -term depends only on u_0 and not on gradients. The temporal oscillations of η are averaged out, yielding a local effective potential.
- (4) In the special case where the fast time dynamics are neglected (i.e., the ∂_τ^2 term in (4.6) is omitted), the corrector χ_k becomes independent of τ , and $\widetilde{\psi}^{\text{eff}}$ reduces to the local form:

$$\widetilde{\psi}_{\text{local}}^{\text{eff}}(\nabla u_0) = \iiint_{Y \times Z^2} \psi \left(y, \tau, \zeta, \nabla u_0 + \sum_{k=1}^N \nabla_y \chi_k(y) \partial_{x_k} u_0 \right) dy d\tau d\zeta.$$

- (5) The derivation above assumes that the cell problem (4.6) is well-posed. This follows from the Fredholm alternative, since the right-hand side has zero mean in y :

$$\int_Y \sum_{i=1}^N \frac{\partial a_{ik}}{\partial y_i}(y) dy = 0.$$

- (6) The nonlocal nature of $\widetilde{\psi}^{\text{eff}}$ is the main novelty of our homogenization result: the separated, temporally oscillating nonlinear mechanisms generate effective macroscopic behavior with memory. This memory effect cannot be captured by standard homogenization frameworks that treat all nonlinearities as a single monolithic operator.

5. Conclusion

This work has overcome a fundamental limitation in homogenization theory by developing the first rigorous framework for nonlinear hyperbolic equations with separated nonlinear mechanisms operating at distinct temporal scales. Our main result, Theorem 4.1, establishes convergence to an effective macroscopic model featuring nonlocal dissipation arising from the fast temporal oscillations, capturing the emergent interplay between amplitude-dependent

potential and gradient-dependent forcing. This provides a theoretical foundation for wave propagation in heterogeneous media with multiple relaxation processes, with applications ranging from seismic analysis in hydrocarbon reservoirs to ultrasonic characterization of aerospace composites.

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Declarations

Conflict of Interest The authors declare that there are no conflict of interest regarding the publication of this paper.

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